

統計学 C 第 8 回分析レポート

JLSPY-J データ (PY140) を用いた重回帰分析・ダミー変数・交互作用

2026-01-14

```
# データの読み込み (csv, sav, dta いずれかに対応)
if (file.exists("PY140.csv")) {
  data <- read_csv("PY140.csv")
} else if (file.exists("PY140.sav")) {
  data <- read_sav("PY140.sav")
} else if (file.exists("PY140.dta")) {
  data <- read_dta("PY140.dta")
} else {
  stop(" データファイル (PY140) が見つかりません。")
}

# --- 変数の作成 ---

# 1. 女性ダミーの作成 (women_d)
# 元の sex 変数: 1=男性, 2=女性
# 変換後: 0=男性, 1=女性
data$women_d <- car::recode(as.numeric(data$sex), '1=0; 2=1')
data$women_d <- as.factor(data$women_d)

# 2. 教育年数 (reseduy)
# ZQ23A を教育年数に換算
data$reseduy <- car::recode(as.numeric(data$ZQ23A),
                           "1=9; 2=12; 3=14; 4=14; 5=16; 6=18; 7=NA; 9=NA")

# 3. 年収 (resincome)
# ZQ47A(カテゴリ) を金額 (中央値) に換算
data$resincome <- car::recode(as.numeric(data$ZQ47A),
                              '1=0; 2=12.5; 3=50; 4=100; 5=200; 6=300; 7=400;
                              8=500; 9=700; 10=1000; 11=1500; 12=2000; 13=2500;
                              14=NA; 99=NA')
```

```
# 4. 年齢 (age)
# 生まれ年 (ybirth) から 2007年時点の年齢を計算
```

```
data$age <- data$ybirth
data$age <- 2007 - data$age
```

```
# モデル 3: 年収 ~ 年齢 + 女性ダミー + 教育年数
```

```
regresult3 <- lm(resincome ~ age + women_d + reseduy, data = data)
summary(regresult3)
```

```
##
## Call:
## lm(formula = resincome ~ age + women_d + reseduy, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -441.11 -115.93   -2.31    99.11 2149.46
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -311.8921    35.9665  -8.672  < 2e-16 ***
## age          15.0402     0.7564  19.883  < 2e-16 ***
## women_d1    -133.1390     6.5959 -20.185  < 2e-16 ***
## reseduy       12.5885     1.7972   7.004 3.03e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 183.4 on 3106 degrees of freedom
## (257 observations deleted due to missingness)
## Multiple R-squared:  0.2203, Adjusted R-squared:  0.2196
## F-statistic: 292.6 on 3 and 3106 DF,  p-value: < 2.2e-16
```

```
# モデル 4: 年齢の 2乗項を追加
```

```
regresult4 <- lm(resincome ~ age + I(age^2) + women_d + reseduy, data = data)
summary(regresult4)
```

```
##
## Call:
## lm(formula = resincome ~ age + I(age^2) + women_d + reseduy,
##      data = data)
##
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -409.21 -117.58   -2.06   95.43 2165.75
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1265.0125   158.5965  -7.976 2.10e-15 ***
## age          84.2538    11.2456   7.492 8.78e-14 ***
## I(age^2)     -1.2247     0.1985  -6.169 7.79e-10 ***
## women_d1    -133.6793     6.5575 -20.386 < 2e-16 ***
## reseduy      12.5357     1.7866   7.017 2.78e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 182.3 on 3105 degrees of freedom
## (257 observations deleted due to missingness)
## Multiple R-squared:  0.2298, Adjusted R-squared:  0.2288
## F-statistic: 231.6 on 4 and 3105 DF,  p-value: < 2.2e-16
```

```
vif(regresult4)
```

```
##      age      I(age^2)      women_d      reseduy
## 226.477678 226.494311   1.006235   1.015721
```

```
# 学歴ダミーの作成 (resedu_d)
# 基準 (0): 高卒 (2)
# ダミー 1: 中卒 (1)
# ダミー 2: 専門・短大・高専 (3,4)
# ダミー 3: 大卒・院卒 (5,6)
data$resedu_d <- car::recode(as.numeric(data$ZQ23A),
                             '1=1; 2=0; 3=2; 4=2; 5=3; 6=3; 7=NA; 9=NA')
data$resedu_d <- as.factor(data$resedu_d)

# モデル 5: 教育年数の代わりに学歴ダミーを使用
regresult5 <- lm(resincome ~ age + women_d + resedu_d, data = data)
summary(regresult5)
```

```
##
## Call:
## lm(formula = resincome ~ age + women_d + resedu_d, data = data)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -419.20 -117.24   -4.05   95.95 2142.43
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -157.653     23.476  -6.716 2.22e-11 ***
## age          15.154       0.759  19.967 < 2e-16 ***
## women_d1     -133.506       6.772 -19.715 < 2e-16 ***
## resedu_d1     -99.566      27.542  -3.615 0.000305 ***
## resedu_d2      16.691       8.877   1.880 0.060175 .
## resedu_d3      46.479       8.370   5.553 3.04e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 183.3 on 3104 degrees of freedom
## (257 observations deleted due to missingness)
## Multiple R-squared:  0.2217, Adjusted R-squared:  0.2205
## F-statistic: 176.8 on 5 and 3104 DF,  p-value: < 2.2e-16
```

```
# 無職ダミーの作成 (nojob_d)
# ZQ03: 1=仕事あり (0), 2=仕事なし (1)
data$nojob_d <- car::recode(as.numeric(data$ZQ03), '1=0; 2=1')
data$nojob_d <- as.factor(data$nojob_d)
```

```
# クロス集計で確認
table(性別=data$women_d, 無業=data$nojob_d)
```

```
##      無業
## 性別    0    1
##    0 1514  179
##    1 1248  426
```

```
# モデル 6: 無職ダミーを追加
regresult6 <- lm(resincome ~ age + women_d + nojob_d + resedu_d, data = data)
summary(regresult6)
```

```
##
## Call:
## lm(formula = resincome ~ age + women_d + nojob_d + resedu_d,
##     data = data)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -380.85  -91.01  -11.82   76.92 2123.45
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -128.9924    21.3048  -6.055 1.58e-09 ***
## age          14.8688     0.6879   21.614 < 2e-16 ***
## women_d1    -101.1687     6.2621 -16.156 < 2e-16 ***
## nojob_d1    -207.4075     7.9780 -25.997 < 2e-16 ***
## resedu_d1    -73.2725    24.9823  -2.933 0.00338 **
## resedu_d2     15.6275     8.0457   1.942 0.05218 .
## resedu_d3     46.5367     7.5859   6.135 9.62e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 166.1 on 3103 degrees of freedom
## (257 observations deleted due to missingness)
## Multiple R-squared:  0.3609, Adjusted R-squared:  0.3597
## F-statistic: 292.1 on 6 and 3103 DF,  p-value: < 2.2e-16
```

```
# 男性のみ (women_d == 0)
print("--- 男性のみの結果 ---")
```

```
## [1] "--- 男性のみの結果 ---"
```

```
regresult7 <- lm(resincome ~ age + nojob_d + resedu_d,
                 data = data, subset = (women_d == 0))
summary(regresult7)
```

```
##
## Call:
## lm(formula = resincome ~ age + nojob_d + resedu_d, data = data,
##     subset = (women_d == 0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -434.95  -99.74   -9.86   75.84 2075.84
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -367.817      33.962 -10.830 < 2e-16 ***
## age          23.293       1.103  21.110 < 2e-16 ***
## nojob_d1     -212.053      16.063 -13.201 < 2e-16 ***
## resedu_d1    -109.798      34.444  -3.188 0.001462 **
## resedu_d2      7.382       13.493   0.547 0.584371
## resedu_d3     43.187       11.226   3.847 0.000124 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 184.9 on 1553 degrees of freedom
## (134 observations deleted due to missingness)
## Multiple R-squared:  0.3348, Adjusted R-squared:  0.3327
## F-statistic: 156.4 on 5 and 1553 DF, p-value: < 2.2e-16
```

```
# 女性のみ (women_d == 1)
print("--- 女性のみの結果 ---")
```

```
## [1] "--- 女性のみの結果 ---"
```

```
regresult8 <- lm(resincome ~ age + nojob_d + resedu_d,
                  data = data, subset = (women_d == 1))
summary(regresult8)

##
## Call:
## lm(formula = resincome ~ age + nojob_d + resedu_d, data = data,
##     subset = (women_d == 1))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -257.33  -77.59  -14.37   66.46 1787.15
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   10.142     23.983   0.423 0.672427
## age           5.876      0.798   7.363 2.90e-13 ***
## nojob_d1     -180.848      7.933 -22.796 < 2e-16 ***
## resedu_d1     -38.656     34.405  -1.124 0.261377
## resedu_d2      32.300      8.883   3.636 0.000286 ***
## resedu_d3      54.031      9.369   5.767 9.73e-09 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 134.3 on 1545 degrees of freedom
## (123 observations deleted due to missingness)
## Multiple R-squared:  0.2797, Adjusted R-squared:  0.2774
## F-statistic: 120 on 5 and 1545 DF,  p-value: < 2.2e-16

# モデル 9: 年齢 × 女性ダミー の交互作用を追加
regresult9 <- lm(resincome ~ age + women_d + nojob_d + resedu_d + women_d:age, data = data)
summary(regresult9)

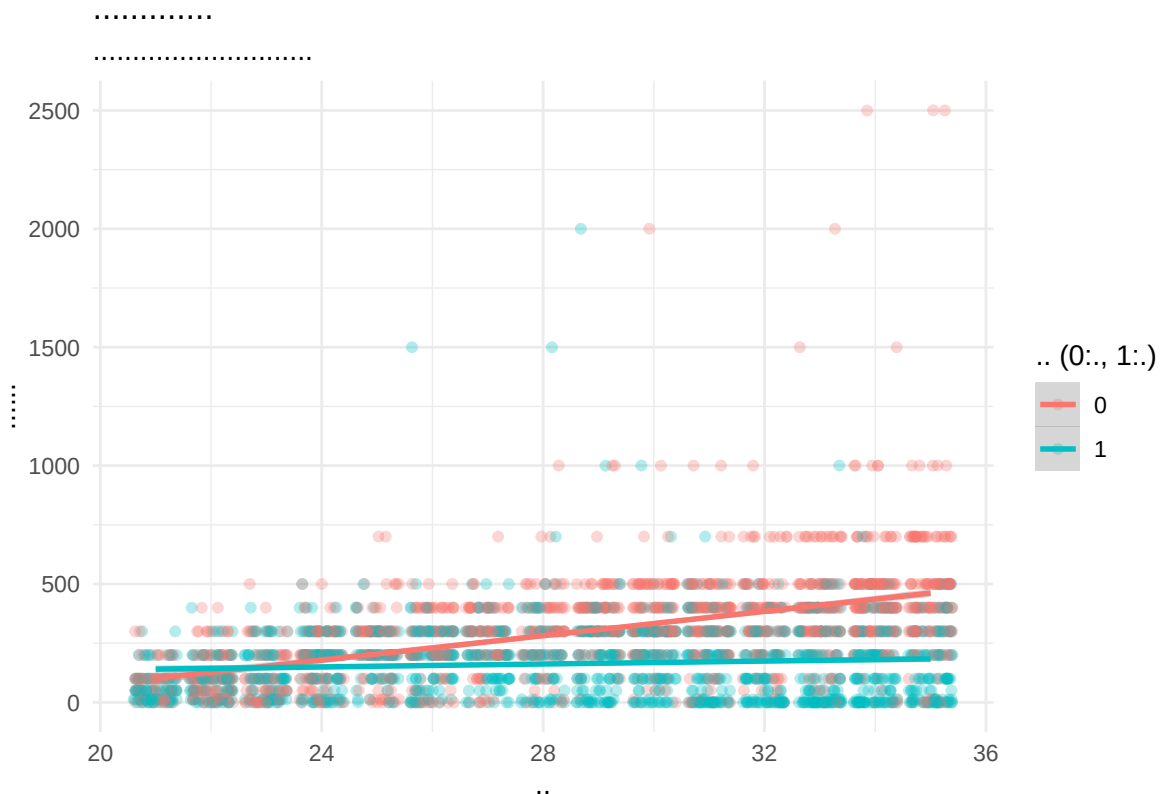
##
## Call:
## lm(formula = resincome ~ age + women_d + nojob_d + resedu_d +
##      women_d:age, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -429.26  -89.63  -14.38   70.56 2081.83
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -383.9235    28.5384  -13.453  < 2e-16 ***
## age           23.5910     0.9478   24.891  < 2e-16 ***
## women_d1      399.3498    38.9488   10.253  < 2e-16 ***
## nojob_d1     -190.4998     7.8780  -24.181  < 2e-16 ***
## resedu_d1     -84.4682    24.3464   -3.469 0.000529 ***
## resedu_d2      20.8009     7.8461    2.651 0.008063 **
## resedu_d3      47.2342     7.3884    6.393 1.87e-10 ***
## age:women_d1  -17.5450     1.3485  -13.011  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 161.8 on 3102 degrees of freedom
## (257 observations deleted due to missingness)
## Multiple R-squared:  0.394, Adjusted R-squared:  0.3926
## F-statistic: 288.1 on 7 and 3102 DF,  p-value: < 2.2e-16
```

```
vif(regresult9, type = 'predictor')
```

```
##              GVIF Df GVIF^(1/(2*Df)) Interacts With      Other Predictors
## age          1.160270 3          1.025085          women_d    nojob_d, resedu_d
## women_d      1.160270 3          1.025085              age    nojob_d, resedu_d
## nojob_d      1.073754 1          1.036221             -- age, women_d, resedu_d
## resedu_d     1.086185 3          1.013874             -- age, women_d, nojob_d
```

```
# 予測値をプロットして傾向を確認
```

```
ggplot(data %>% filter(!is.na(resincome) & !is.na(age) & !is.na(women_d)),
  aes(x = age, y = resincome, color = women_d)) +
  geom_jitter(alpha = 0.3, height = 0.5) + # 実データ (散布図)
  geom_smooth(method = "lm", se = TRUE) + # 回帰直線
  labs(title = "年齢と年収の関係 (男女別)",
    subtitle = "交互作用の可視化：女性（赤）の方が傾きが緩やかであるか？",
    x = "年齢", y = "年収（万円）",
    color = "性別（0: 男, 1: 女）" +
  theme_minimal()
```



```
# 考察
```

男性の年収は年齢とともに急上昇 (+23.6) する年功序列型なのに対し、女性の年収は年齢による上昇幅が極めて小さく (傾き約 6.0)、ほぼフラットに近い。

この結果は、女性が結婚・出産等のライフイベントによりキャリア形成や昇進の機会を逸しやすいこと、あるいは企業側が女性を補助的な業務に固定しがちであるという統計的差別や職域分離の存在を強く示唆している。

最終課題は計量経済学に関連するテーマにしようかなと思っています。